

Inferring causal networks for genomic trios with high-dimensional confounding variables

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16 **Abstract**

17 Causal structure inference under the principle of Mendelian randomization (PMR) enables
18 the study of causal relationships between two molecular phenotypes (e.g., gene expression) by
19 leveraging genetic variants as instrumental variables to impose constraints on edge direction. In
20 practice, inference is often affected by numerous confounding variables, such as other genes,
21 other processes and demographic variables, making it essential to robustly account for con-
22 founders. However, existing methods have limited ability to accommodate high-dimensional
23 confounding. Here, we introduce Mendelian Randomization for Genomic Networks (MRGN),
24 a regression-based method under the PMR for inferring causal structures in genomic trios
25 from individual-level data while accounting for many confounders. MRGN extends beyond
26 the canonical mediation model to distinguish among multiple causal relationships. By prior-
27 itizing conditional independence tests given the genetic variant and confounders, MRGN can
28 detect causal relationships between phenotypes even when their marginal association is weak.
29 Through extensive simulations, we demonstrate improved inference accuracy, robustness, and
30 computational efficiency over existing approaches. We further apply MRGN to data from the
31 Genotype-Tissue Expression (GTEx) Consortium to investigate how genetic variants may reg-
32 ulate gene expression in cis and trans.

33 **Keywords**

34 principle of Mendelian randomization, causal network inference, causal structure learning,
35 confounding, gene regulatory networks

36 **Key Points**

- 37 • Mendelian Randomization for Genomic Networks (MRGN) is a regression-based method
38 for inferring causal structures in genomic trios from individual-level data while account-
39 ing for many confounders. A genomic trio consists of two molecular phenotypes and a
40 genetic variant.
- 41 • MRGN leverages the Principle of Mendelian randomization (PMR), which uses genetic
42 variants as instrumental variables to impose constraints on edge directions in the infer-
43 ence of causal network structure.
- 44 • By prioritizing conditional independence tests given the genetic variant and confounders,
45 MRGN can detect causal relationships between phenotypes even when their marginal
46 association is weak.

47 **1 Introduction**

48 An important goal in biology is to understand how genes regulate one another and how genetic
49 variation influences gene expression [1, 2, 3, 4]. Gene regulation involves complex networks of
50 high dimensions, which may be represented by directed networks where directed edges repre-
51 sent causal (or regulatory) relationships. Although the ultimate goal in biology is to reconstruct
52 gene regulatory networks at the genome level, it is often of interest to understand the causal re-
53 lationship between a molecular phenotype (e.g. the expression level) of two genes, or between
54 two molecular phenotypes (e.g., between gene expression and DNA methylation), which may
55 be influenced by many other variables such as other genes, other biological processes, demo-
56 graphic variables, etc.

57 To learn the (directed) causal relationship between two phenotypes, we can apply the prin-
58 ciple of Mendelian randomization (PMR) and leverage the information in genetic variants. The

59 PMR assumes that genotypes in a natural population are randomly assigned, thus observational
60 omics data can be viewed as being collected from a randomized experiment [5, 6]. This implies
61 that an edge connecting a genotype to a phenotype will always point to the phenotype, but not
62 the other way around [7, 8]. As a result, this constraint on edge direction makes it possible to
63 break the symmetry of association and draw inferences about the regulatory structure among
64 phenotypes [6, 8, 9].

65 Methods based on the PMR framework for causal structure learning differ fundamentally
66 from classical Mendelian randomization (MR) methods. Traditional MR focuses on estimating
67 causal effects and depends on three key assumptions [6, 10, 11]: (i) relevance, where the
68 genetic variant is associated with the molecular phenotype; (ii) independence, where the variant
69 is independent of unmeasured confounders due to random allele segregation; and (iii) exclusion
70 restriction, where the variant influences downstream phenotypes only through the intermediate
71 phenotype, and not through other pathways. In contrast, our PMR-based framework is not
72 designed to estimate causal effect sizes, and does not require the exclusion restriction in the
73 same form as classical MR. Instead, it uses the genetic variant primarily to impose directional
74 constraints on the causal graph, such that edges from molecular phenotypes to genetic variants
75 are not allowed. For the variant to provide useful directional information, it must be associated
76 with at least one phenotype and independent of unmeasured confounders. However, the goal is
77 not to estimate its causal effect on the phenotype, but to leverage this association as a structural
78 anchor for learning the relationship between the two phenotypes. Consequently, the variant
79 does not need to be causal itself; a tag variant can serve as the anchor as long as it provides the
80 relevant association.

81 Many methods have been developed under assumptions similar to the PMR [12, 13, 14,
82 15, 8, 16, 17, 18], although some of these methods are interested mostly in the mediation
83 model $V \rightarrow T_1 \rightarrow T_2$, where V is the genetic variant, T_1 and T_2 are the variables of interest.
84 In this model, the mediator T_1 relays the regulatory information from V to the outcome T_2 ;

85 for example, a genetic variant regulates the expression level of a transcription factor, and its
 86 reduced expression leads to its downstream target gene being under-expressed. However, other
 87 causal relationships may exist among the three variables.

88 We previously developed MRPC, a constraint-based causal structure learning method that
 89 uses patterns of marginal and conditional independence to distinguish five mutually exclusive
 90 basic causal models under the PMR (Figure 1) [8, 19, 20]. They include a null model ($V \rightarrow T_1$
 91 and no relationship between T_1 and T_2 , denoted M0) and four non-null models: mediation
 92 (denoted M1), v-structure (or reverse causation; $V \rightarrow T_1 \leftarrow T_2$, denoted M2), conditional
 93 independence (or horizontal pleiotropy; $T_1 \leftarrow V \rightarrow T_2$, denoted M3), and a fully connected
 94 model ($T_1 \leftarrow V \rightarrow T_2$ and $T_1 \leftrightarrow T_2$, denoted M4). The true model behind M4 may be
 95 directed: the edge between T_1 and T_2 is $T_1 \rightarrow T_2$ or $T_1 \leftarrow T_2$, but these two directions result
 96 in the same likelihood for the trio and are therefore Markov equivalent. Without additional
 97 information, they are indistinguishable and only a bidirected/undirected edge can be inferred
 98 for M4. The primary goal of inference in this framework is the presence and direction of the
 99 $T_1 - T_2$ relationship, while V helps distinguish among competing causal models.

100 The PMR and the five basic causal models require that no confounding variables can ex-
 101 plain away the directed edge between T_1 and T_2 , since this edge is our main interest. How-
 102 ever, as mentioned earlier, many variables can influence the inference due to the complexity
 103 of gene regulation. A confounding variable may influence the inference in different ways.
 104 Yang et al. [14] proposed three types of confounding C for the $T_1 - T_2$ edge: as a con-
 105 founder ($T_1 \leftarrow C \rightarrow T_2$), an intermediate variable ($T_1 \rightarrow C \rightarrow T_2$), or a common child
 106 ($T_1 \rightarrow C \leftarrow T_2$). Specifically, a confounder is a common parent to both T_1 and T_2 and can
 107 induce a false edge between T_1 and T_2 when unaccounted for. This is similar to T_1 and T_2
 108 being marginally dependent in M3. An intermediate variable acts like a mediator between T_1
 109 and T_2 and can potentially explain away a true $T_1 \rightarrow T_2$ edge when this intermediate variable is
 110 accounted for in the inference, leading to a false negative conclusion. By contrast, a common

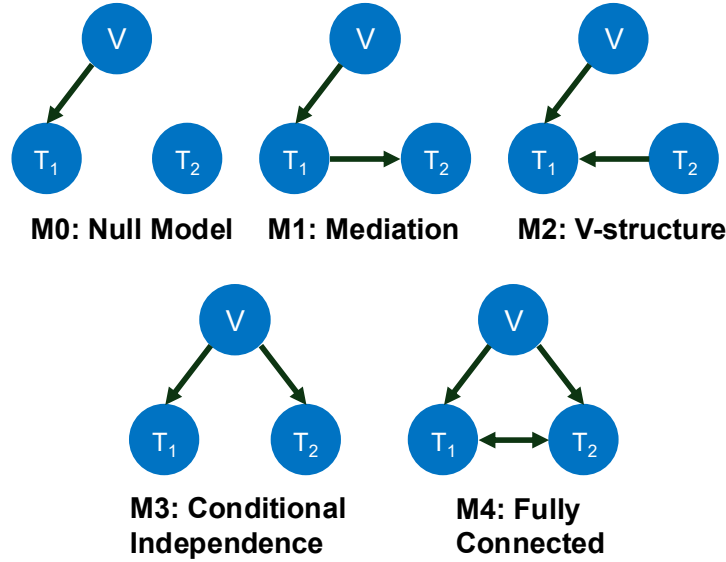


Figure 1: The five basic causal models for a genomic trio that may be inferred under the principle of Mendelian randomization. A genomic trio consists of a genetic variant V and two molecular phenotypes T_1 and T_2 . This figure is adapted from [8].

Alt text: Five three-node graphs ranging from no relationship between the phenotypes to mediation, v-structure, conditional independence, and a fully connected structure, with arrows constrained to point away from the genetic variant.

child can induce a false edge between T_1 and T_2 when accounted for, leading to a false positive conclusion. This is because when the common child is accounted for in the inference, T_1 and T_2 become conditionally dependent, similar to their relationship in M2.

Accounting for confounding variables is therefore crucial in causal network inference [21, 14, 22, 23, 24]. Two basic approaches have emerged to address latent confounding between an exposure and outcome. In the first approach, candidate confounding variables are incorporated into the graph as additional nodes in the network; MRPC and MR-GGI, for example, can infer larger networks [8, 19, 18], but the computational complexity of network inference grows super-exponentially with the number of nodes. The second approach focuses on identifying and accounting for confounders [21, 14, 22]. This approach includes all the confounders as a whole in the tests for edge presence, and does not increase the size of the network. For example, we can include the confounders when testing the $T_1 - T_2$ edge in a trio. The trio remains a network of three nodes, but the tests are conditioned on these confounders. However, these

124 methods focus on the mediation model.

125 Herein we present Mendelian Randomization for Genomic Networks (MRGN), a constraint-
126 based method for inferring the causal structure of a genomic trio from individual-level data in
127 the presence of potentially hundreds of confounders. Like MRPC, MRGN is also a constraint-
128 based method such as the classic PC algorithm (named after its developers Peter Spirtes and
129 Clark Glymour) [25], inferring graph structure from marginal and conditional independence
130 tests. While MRGN retains from MRPC the PMR constraint and the five basic causal models,
131 it differs from MRPC in two important aspects. First, MRGN prioritizes conditional inde-
132 pendence tests over marginal independence tests, preserving $T_1 - T_2$ relationships that may
133 be weakened or obscured when genotypes are combined due to Simpson’s paradox. Sec-
134 ond, rather than incorporating candidate confounders as additional network nodes, MRGN
135 identifies confounders and adjusts for them as a whole in the independence tests. This al-
136 lows large numbers of confounders to be accommodated without increasing the dimension
137 of the inferred network and substantially reduces computational complexity. We evaluate
138 these methodological changes through extensive simulations and apply MRGN to study cis-
139 and trans-regulatory relationships using the whole blood data from the Genotype-Tissue Ex-
140 pression (GTEx) Consortium [26]. [MRGN provides statistical evidence for potential causal](#)
141 [relationships, and additional experimental validation or perturbation studies may be needed to](#)
142 [establish causality more definitively.](#)

143 **2 Methods**

144 **2.1 The MRGN method**

145 The five causal models for a trio can each be described by a set of conditional and marginal
146 dependence relations (Figure 2a; Table 1), which may be encoded in the coefficients of two
147 regression models that account for confounders, assuming that T_1 and T_2 are centered so that

the intercept is zero:

$$T_1 = \beta_{11}V + \beta_{12}T_2 + \mathbf{\Gamma}_1\mathbf{U} + \epsilon_1; \quad (1)$$

$$T_2 = \beta_{21}V + \beta_{22}T_1 + \mathbf{\Gamma}_2\mathbf{U} + \epsilon_2, \quad (2)$$

where β s are coefficients of interest, vector $\mathbf{\Gamma}_i$ the coefficients for the confounder vector $\mathbf{U}_i$, and ϵ_i the random error that follows a normal distribution with mean 0 and standard deviation σ_i , where $i = 1, 2$.

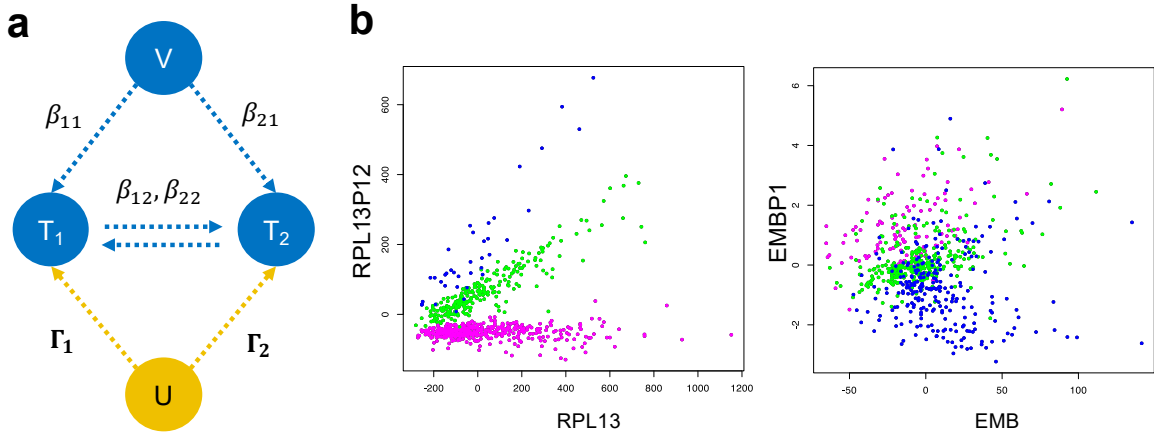


Figure 2: The MRGN method. (a) A genomic trio includes a genetic variant (V), two molecular phenotypes (T_1 and T_2), and a set of confounders ($\mathbf{U}$). (b) Two examples of Simpson's paradox in genomic trios from the GTEx whole blood data. Each dot represents an individual, with colors indicating genotypes. Ignoring genotype, the association is weakened in the left example and disappears in the right example.

Alt text: Panel a shows a causal graph with a genetic variant, two molecular phenotypes, and multiple confounders. Panel b shows two scatterplots illustrating Simpson's paradox, with the dot color-coded by genotype and within-genotype associations differing from the overall associations.

Our conditional independence tests determine whether each β is zero, with tests of β_{12} and β_{21} being equivalent because they represent the same conditional dependence. For example, in the mediation model $V \rightarrow T_1 \rightarrow T_2$, V and T_2 are conditionally independent given T_1 , corresponding to $\beta_{21} = 0$, while the other coefficients are nonzero. Models M0 and M3 can

156 similarly be distinguished using conditional independence tests alone.

157 In the v-structure M2, for example $V \rightarrow T_1 \leftarrow T_2$, V and T_2 are marginally independent but
158 conditionally dependent given T_1 . Thus, all conditional associations are nonzero in both M2
159 and the fully connected model M4, and these models are distinguished by marginal dependence
160 between V and T_2 .

161 MRGN prioritizes conditional independence tests because the $T_1 - T_2$ association within
162 genotypes may weaken or disappear when genotypes are combined, as in Simpson’s Paradox
163 (Figure 2b). In contrast, MRPC tests marginal independence first and permanently removes a
164 $T_1 - T_2$ edge after a negative marginal test, potentially reducing power.

165 MRGN performs the standard t tests on the regression coefficients for conditional depen-
166 dence and the standard test on Pearson correlation for marginal dependence. Since at most
167 five tests are performed for the edges of interest, we set the significance level for each test to
168 0.01, providing Bonferroni control of the familywise type I error rate at 0.05 for these tests,
169 conditional on the selected confounders. When the data is not informative enough for any of
170 the basic models to be inferred, MRGN categorizes the trio as “other”.

171 2.2 Confounder selection

172 We perform confounder selection before causal network inference, using a two-step procedure
173 similar to that in the GMAC method [14]. First, we identify candidate confounders from a
174 pool of potential confounding variables. Confounders, which are defined as parents of T_1 and
175 T_2 , should be independent of V . We therefore exclude variables that are marginally associated
176 with V . Next, we select confounders based on their statistical association with either T_1 or T_2 .
177 Similar to [14], we test whether the regression of both T_1 and T_2 on a candidate confounder is
178 statistically significant. Both steps allow for an additional step of the q -value method to control
179 the false discovery rate (FDR) [27].

180 In our R implementation, we substantially improve computational efficiency in two key

Table 1: Conditional and marginal independence tests in MRGN. Regression coefficients β_{11} and β_{12} (from Equation 1) and β_{21} and β_{22} (from Equation 2) are used to test conditional independence. The conditional test for the $T_1 - T_2$ pair can be tested by β_{12} or equivalently β_{22} , reducing the number of distinct conditional tests to three. Marginal independence is tested using Pearson correlations r_{V,T_1} and r_{V,T_2} . The first three models each have two possible configurations.

		Conditional Test			Marginal Test	
Null Hypothesis		$T_1 \perp\!\!\!\perp V$	$T_1 \perp\!\!\!\perp T_2$	$T_2 \perp\!\!\!\perp V$	$V \perp\!\!\!\perp T_1$	$V \perp\!\!\!\perp T_2$
Conditioning Set		$T_2, \mathbf{U}$	$V, \mathbf{U}$	$T_1, \mathbf{U}$	–	–
Parameter		β_{11}	β_{12} (or β_{22})	β_{21}	r_{V,T_1}	r_{V,T_2}
Model	Configuration					
M0	$V \rightarrow T_1; T_2$ is singleton	$\neq 0$	$= 0$	$= 0$	–	–
	$V \rightarrow T_2; T_1$ is singleton	$= 0$	$= 0$	$\neq 0$	–	–
M1	$V \rightarrow T_1 \rightarrow T_2$	$\neq 0$	$\neq 0$	$= 0$	–	–
	$V \rightarrow T_2 \rightarrow T_1$	$= 0$	$\neq 0$	$\neq 0$	–	–
M2	$V \rightarrow T_1 \leftarrow T_2$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$= 0$
	$V \rightarrow T_2 \leftarrow T_1$	$\neq 0$	$\neq 0$	$\neq 0$	$= 0$	$\neq 0$
M3	$T_1 \leftarrow V \rightarrow T_2$	$\neq 0$	$= 0$	$\neq 0$	–	–
M4	$T_1 \leftarrow V \rightarrow T_2; T_1 \longleftrightarrow T_2$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$

ways. First, we implemented parallelization in both steps, allowing for faster evaluation when the pool of confounders is large. Second, we used the R package *propagate* for correlation matrix computation by partitioning the problem into smaller, memory-efficient blocks [28], which scales more efficiently to large genomic datasets.

The overall MRGN analysis workflow, from data preprocessing through confounder identification and selection to causal structure inference, is summarized in Supplementary Figure 1.

2.3 Simulation study

Simulation setup

Each simulated trio consists of a bi-allelic genetic variant V with three genotypes and two genes (T_1, T_2) connected under one of the five causal models. Nodes in a trio are generated in topological order. The genotype V_i is simulated first under Hardy–Weinberg equilibrium with

192 the minor allele frequency θ :

$$\Pr(V_i = 0) = (1 - \theta)^2, \quad \Pr(V_i = 1) = 2\theta(1 - \theta), \quad \Pr(V_i = 2) = \theta^2.$$

193 Each remaining node Y_j (a gene or a confounder) is sampled from

$$Y_j \sim N\left(\sum_{i \in S_j} \beta_i Y_i, \sigma^2\right),$$

194 where S_j is the parent set of Y_j , which may include V , the other gene, or confounders. For M4,
195 the true edge between T_1 and T_2 can be in either direction. For simplicity, we fix all M4 trios
196 to the configuration $T_1 \leftarrow V \rightarrow T_2$ with $T_1 \rightarrow T_2$.

197 We performed two simulations. Simulation A evaluated confounder selection and causal
198 network inference across three effect sizes and four sample sizes. Effect sizes were defined by
199 the effects of V on T and between the two T s:

- 200 • Weak: Uniform (0.1, 0.5) and Uniform (0.1, 0.3), respectively;
- 201 • Moderate: Uniform (0.5, 1) and Uniform (0.3, 0.5), respectively;
- 202 • Strong: Uniform (1, 1.5) and Uniform (0.5, 1), respectively.

203 The weak setting approximated the genotype–cis gene associations in the GTEx whole-blood
204 data [26]: the median partial correlation between V and T_1 , conditional on the true con-
205 founders, was 0.14 in the simulations, compared with a mode of approximately 0.13 for the
206 corresponding partial correlations conditional on selected PCs in GTEx. The moderate and
207 strong settings were consistent with those used by [14].

208 Each trio included 1–50 confounders, one intermediate, and one common child. Con-
209 founder effects were sampled from Uniform (0, 0.3), intermediate effects from Uniform (0.05,
210 0.5), and common-child effects from Uniform (1, 1.5), each with a 50% probability of being

negative. Confounder effects were also calibrated to expression-derived PCs in GTEx, with median absolute correlations of 0.10 in the simulated and GTEx data.

We considered sample sizes of 50, 150, 300, and 1000, fixed the residual variance at 1, and sampled the minor allele frequency from $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. For each combination of effect size and sample size, we generated 20 datasets for each of the five causal models, yielding $3 \times 4 \times 5 \times 20 = 1,200$ trios.

Simulation B evaluated the impact of different types of confounding using four scenarios: confounders only, confounders plus one intermediate, confounders plus one common child, and all three types. We considered the same three effect sizes but fixed the sample size at 670, matching the GTEx v8 whole-blood data used in our application. For each scenario, we generated 20 datasets for each causal model and effect size, yielding $3 \times 5 \times 20 = 300$ trios.

Performance metrics

We evaluated three sets of recall (i.e., power) and precision to quantify the performance on the entire model of a trio, on individual edges and specifically on the $T_1 - T_2$ edge.

1. On the causal model type: a trio is considered correctly inferred when its inferred model type matches its true type, thus requiring all edges to be inferred correctly.

$$\text{Recall}_{\text{model M}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios of the causal model M}}; \quad (3)$$

$$\text{Precision}_{\text{model M}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios inferred to be of the causal model M}}. \quad (4)$$

2. On individual edges: we first calculated recall and precision based on individual edges in a single trio. Similar to our earlier work [8], we counted an edge with the wrong

229 direction as a half correct edge.

$$\text{Recall}_{\text{edges}} = \frac{\text{Number of correct edges in inferred graph}}{\text{Total edges in true graph}}; \quad (5)$$

$$\text{Precision}_{\text{edges}} = \frac{\text{Number of correct edges In inferred graph}}{\text{Total edges in inferred graph}}. \quad (6)$$

230 We then calculated the mean metric across all trios.

231 3. On the $T_1 - T_2$ edge (allowing for comparison with the GMAC method): the direct of
232 this edge was ignored for this evaluation. Across all trios,

$$\text{Recall}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with a true } T_1 - T_2 \text{ edge}}; \quad (7)$$

$$\text{Precision}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with an inferred } T_1 - T_2 \text{ edge}}. \quad (8)$$

233 3 Results

234 3.1 Simulation Results

235 MRGN improves power and precision to detect edges in the presence of confounding

236 Before applying MRGN, we examined two approaches to confounder selection: controlling the
237 FDR at 5% with the q -value method (denote “CS- q ”), and not correcting for multiple testing
238 but instead setting the type I error rate of each individual test at $\alpha < 0.01$ (denoted “CS- α ”).
239 We evaluated their ability to identify true confounders using precision and recall. Selection
240 performance depended strongly on sample size but little on the simulated effect sizes (Supple-
241 mentary Figure 2; Supplementary Figure 3). CS- q was highly conservative at small sample
242 sizes, with near-zero recall at $n = 50$ and $n = 150$, but improved substantially with increas-
243 ing sample size, reaching a precision of 0.84 and recall of 0.79 at $n = 1,000$. In contrast, as

244 expected, CS- α selected many more covariates and consequently achieved substantially higher
245 recall, increasing from 0.08 at $n = 50$ to 0.93 at $n = 1,000$, at the cost of low precision
246 (0.02–0.21). Thus, CS-q and CS- α represent different precision-recall tradeoffs: CS-q favors
247 a smaller, more specific set of confounders, whereas CS- α favors more complete confounder
248 recovery at the expense of including many non-confounders.

249 Simulation A showed that MRGN reliably detected the $T_1 - T_2$ edge when sufficient in-
250 formation was available, whereas recovery of the exact causal model was more challenging
251 and varied across models (Table 2; Table 3). Increasing the effect size or sample size gener-
252 ally improved MRGN performance. With CS-q, recall for the $T_1 - T_2$ edge increased from
253 0.257 under weak effects to 0.830 under strong effects, and from 0.194 at $n = 50$ to 0.844
254 at $n = 1,000$. Recall for individual causal models also generally increased with effect size
255 and sample size. For example, as n increased from 50 to 1,000, recall increased from 0.183 to
256 0.767 for M1 and from 0 to 0.633 for M4, whereas the improvement for M3 was more modest,
257 from 0.150 to 0.333. Precision was generally high (>0.8) for M2 and M3 except under weak
258 effects. MRGN with CS- α showed qualitatively similar trends, generally with somewhat lower
259 recall and higher precision than CS-q. MRPC recovered M0 and M3 substantially better, con-
260 sistent with its conservative behavior in calling edges. MR-GGI detected fewer edges overall
261 and generally recovered M1 and M4 well, but rarely recovered the other models.

262 At the edge level, MRGN generally detected the $T_1 - T_2$ edge with higher recall than MRPC,
263 while its performance relative to GMAC depended on the amount of information available (Ta-
264 ble 2; Table 3). GMAC had substantially higher recall under weak effects and at small sample
265 sizes, but the difference narrowed as effect size and sample size increased, with MRGN main-
266 taining comparable or higher precision. Because the GMAC mediation test is a special case of
267 MRGN, these results suggest that GMAC is more sensitive for detecting a $T_1 - T_2$ relation-
268 ship under limited information, whereas MRGN provides greater resolution of the underlying
269 causal structure.

270 To better illustrate the behavior of MRGN in causal-model inference, we aggregated the
271 confusion matrices across effect sizes and sample sizes (Supplementary Table 1). For MRGN
272 with CS-q, inferred M1 trios were a mixture of all five true models, whereas inferred M4
273 structures consisted mainly of true M3 and M4. These difficulties were not unique to MRGN.
274 Although MRPC was more conservative in assigning causal models, it also had difficulty dis-
275 tinguishing M1 from M4: 51 (54%) of the 95 structures inferred as M1 were true M4.

276 The distinction between M1 and M4 and that between M3 and M4 depend on the outcome
277 of a single conditional independence test. Thus, these two patterns illustrate the limitation of
278 hypothesis testing in constraint-based inference. A true M4 may be inferred as M1 when the
279 null hypothesis is not rejected because of insufficient statistical power. Conversely, a true M3
280 may be inferred as M4 when the null hypothesis of no $T_1 - T_2$ edge is incorrectly rejected.
281 Thus, both insufficient evidence for a true edge and false-positive evidence for an absent edge
282 can affect discrete causal-model assignment.

283 We further evaluated the runtime for all methods on CPUs of a shared computing cluster.
284 MRGN was computationally efficient: confounder selection required 16 minutes in the largest
285 setting examined (1,000 covariates and 1,000 trios), and causal network inference required a
286 median of 0.0056 seconds per trio (Supplementary Figure 4). For causal network inference,
287 MRGN was approximately twice as fast as MR-GGI, six times faster than MRPC, and 100
288 times faster than GMAC (Supplementary Figure 5).

289 **The impact of confounding variables on causal network inference**

290 We examined how errors in confounder selection affected MRGN inference. False positive
291 confounders had relatively little effect on edge precision, although recall varied with the selec-
292 tion method and simulation setting (Supplementary Figure 6). In contrast, increasing numbers
293 of false negative confounders generally reduced edge recall, with the effect being most pro-
294 nounced at smaller sample sizes (Supplementary Figure 7).

Table 2: Mean recall and precision (standard errors in parentheses) from Simulation A, stratified by method and effect size. NA indicates that the metric cannot be evaluated because no trios were inferred as the corresponding model. MRGN (Truth) uses the true confounders in the inference. GMAC assumes a mediation model and only infers the $T_1 - T_2$ edge. Highest mean precision and recall is highlighted in each row (excluding MRGN truth).

Weak Effects																		
	MRGN (Truth)			MRGN (CS-q)			MRGN (CS-a)			MRPC (CS-q)			MR-GGI (CS-q)			GMAC		
	Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision	
M0	0.490 (0.050)	0.605 (0.054)		0.130 (0.034)	0.265 (0.063)		0.130 (0.034)	0.302 (0.070)		0.150 (0.046)	0.300 (0.084)		0.050 (0.022)	0.263 (0.101)		–	–	
M1	0.430 (0.050)	0.462 (0.052)		0.210 (0.041)	0.233 (0.045)		0.170 (0.038)	0.250 (0.053)		0.033 (0.023)	0.500 (0.250)		0.050 (0.022)	0.208 (0.083)		–	–	
M2	0.010 (0.010)	0.125 (0.117)		0.010 (0.010)	0.167 (0.152)		0.010 (0.010)	0.200 (0.179)		0.000 (0.000)	0.000 (0.000)		0.000 (0.000)	NA		–	–	
M3	0.350 (0.048)	0.814 (0.059)		0.070 (0.026)	0.538 (0.138)		0.130 (0.034)	0.765 (0.103)		0.050 (0.028)	0.750 (0.217)		0.010 (0.010)	0.500 (0.354)		–	–	
M4	0.220 (0.041)	0.957 (0.043)		0.180 (0.038)	0.474 (0.081)		0.150 (0.036)	0.556 (0.096)		0.033 (0.023)	1.000 (0.000)		0.300 (0.046)	0.476 (0.063)		–	–	
$T_1 - T_2$ Edge	0.397 (0.028)	0.960 (0.018)		0.257 (0.025)	0.575 (0.043)		0.207 (0.023)	0.620 (0.049)		0.039 (0.014)	1.000 (0.000)		0.083 (0.016)	0.676 (0.077)		0.643 (0.028)	0.581 (0.027)	
Moderate Effects																		
	MRGN (Truth)			MRGN (CS-q)			MRGN (CS-a)			MRPC (CS-q)			MR-GGI (CS-q)			GMAC		
	Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision	
M0	0.740 (0.044)	0.718 (0.044)		0.370 (0.048)	0.463 (0.056)		0.360 (0.048)	0.720 (0.063)		0.517 (0.065)	0.508 (0.064)		0.030 (0.017)	0.600 (0.219)		–	–	
M1	0.710 (0.045)	0.573 (0.044)		0.550 (0.050)	0.355 (0.038)		0.520 (0.050)	0.433 (0.045)		0.133 (0.044)	0.296 (0.088)		0.460 (0.050)	0.605 (0.056)		–	–	
M2	0.420 (0.049)	0.977 (0.023)		0.360 (0.048)	0.923 (0.043)		0.250 (0.043)	1.000 (0.000)		0.183 (0.050)	0.524 (0.109)		0.000 (0.000)	NA		–	–	
M3	0.710 (0.045)	0.947 (0.026)		0.300 (0.046)	0.857 (0.059)		0.350 (0.048)	1.000 (0.000)		0.450 (0.064)	0.771 (0.071)		0.010 (0.010)	0.500 (0.354)		–	–	
M4	0.600 (0.049)	0.923 (0.033)		0.450 (0.050)	0.542 (0.055)		0.380 (0.049)	0.704 (0.062)		0.283 (0.058)	0.850 (0.080)		0.600 (0.049)	0.429 (0.042)		–	–	
$T_1 - T_2$ Edge	0.750 (0.025)	0.970 (0.011)		0.647 (0.028)	0.700 (0.028)		0.543 (0.029)	0.819 (0.027)		0.367 (0.036)	0.971 (0.020)		0.423 (0.029)	0.774 (0.033)		0.850 (0.021)	0.687 (0.024)	
Strong Effects																		
	MRGN (Truth)			MRGN (CS-q)			MRGN (CS-a)			MRPC (CS-q)			MR-GGI (CS-q)			GMAC		
	Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision		Recall	Precision	
M0	0.910 (0.029)	0.901 (0.030)		0.580 (0.049)	0.806 (0.047)		0.580 (0.049)	0.829 (0.045)		0.833 (0.048)	0.704 (0.054)		0.000 (0.000)	NA		–	–	
M1	0.860 (0.035)	0.534 (0.039)		0.800 (0.040)	0.408 (0.035)		0.580 (0.049)	0.443 (0.043)		0.450 (0.064)	0.422 (0.062)		0.810 (0.039)	0.551 (0.041)		–	–	
M2	0.760 (0.043)	1.000 (0.000)		0.660 (0.047)	0.985 (0.015)		0.500 (0.050)	1.000 (0.000)		0.550 (0.064)	0.635 (0.067)		0.000 (0.000)	NA		–	–	
M3	0.860 (0.035)	0.977 (0.016)		0.450 (0.050)	0.918 (0.039)		0.280 (0.045)	1.000 (0.000)		0.650 (0.062)	1.000 (0.000)		0.010 (0.010)	1.000 (0.000)		–	–	
M4	0.290 (0.045)	0.935 (0.044)		0.230 (0.042)	0.354 (0.059)		0.190 (0.039)	0.373 (0.068)		0.083 (0.036)	0.294 (0.111)		0.340 (0.047)	0.256 (0.038)		–	–	
$T_1 - T_2$ Edge	0.880 (0.019)	0.985 (0.007)		0.830 (0.022)	0.759 (0.024)		0.630 (0.028)	0.815 (0.026)		0.700 (0.034)	0.947 (0.019)		0.600 (0.028)	0.709 (0.029)		0.963 (0.011)	0.739 (0.022)	

Table 3: Mean recall and precision (standard errors in parentheses) from Simulation A, stratified by method and sample size. NA indicates that the metric cannot be evaluated because no trios were inferred as the corresponding model. MRGN (Truth) uses the true confounders in the inference. GMAC assumes a mediation model and only infers the $T_1 - T_2$ edge. Highest mean precision and recall is highlighted in each row (excluding MRGN truth).

n=50	MRGN (Truth)		MRGN (CS-q)		MRGN (CS-a)		MRPC (CS-q)		MR-GGI (CS-q)		GMAC	
	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision
M0	0.300 (0.059)	0.353 (0.067)	0.333 (0.061)	0.392 (0.068)	0.000 (0.000)	NA	0.233 (0.055)	0.359 (0.077)	0.000 (0.000)	0.000 (0.000)	-	-
M1	0.117 (0.041)	0.333 (0.103)	0.183 (0.050)	0.314 (0.078)	0.000 (0.000)	NA	0.217 (0.053)	0.433 (0.090)	0.200 (0.052)	0.324 (0.077)	-	-
M2	0.117 (0.041)	1.000 (0.000)	0.067 (0.032)	0.667 (0.192)	0.000 (0.000)	NA	0.100 (0.039)	0.857 (0.132)	0.000 (0.000)	NA	-	-
M3	0.183 (0.050)	0.846 (0.100)	0.150 (0.046)	1.000 (0.000)	0.000 (0.000)	NA	0.167 (0.048)	1.000 (0.000)	0.050 (0.028)	1.000 (0.000)	-	-
M4	0.000 (0.000)	NA	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	NA	0.000 (0.000)	NA	0.017 (0.017)	0.053 (0.051)	-	-
$T_1 - T_2$ Edge	0.156 (0.027)	1.000 (0.000)	0.194 (0.029)	0.814 (0.059)	0.000 (0.000)	NA	0.206 (0.030)	1.000 (0.000)	0.161 (0.027)	0.690 (0.071)	0.606 (0.036)	0.736 (0.036)
n=150												
n=150	MRGN (Truth)		MRGN (CS-q)		MRGN (CS-a)		MRPC (CS-q)		MR-GGI (CS-q)		GMAC	
	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision
M0	0.733 (0.057)	0.710 (0.058)	0.300 (0.059)	0.500 (0.083)	0.233 (0.055)	0.424 (0.086)	0.600 (0.063)	0.679 (0.064)	0.033 (0.023)	0.400 (0.219)	-	-
M1	0.600 (0.063)	0.400 (0.052)	0.317 (0.060)	0.211 (0.043)	0.067 (0.032)	0.200 (0.089)	0.267 (0.057)	0.432 (0.081)	0.333 (0.061)	0.408 (0.070)	-	-
M2	0.250 (0.056)	1.000 (0.000)	0.183 (0.050)	0.917 (0.080)	0.017 (0.017)	1.000 (0.000)	0.300 (0.059)	0.621 (0.090)	0.000 (0.000)	NA	-	-
M3	0.617 (0.063)	0.902 (0.046)	0.267 (0.057)	1.000 (0.000)	0.017 (0.017)	1.000 (0.000)	0.483 (0.065)	0.906 (0.052)	0.000 (0.000)	NA	-	-
M4	0.117 (0.041)	1.000 (0.000)	0.067 (0.032)	0.211 (0.094)	0.000 (0.000)	NA	0.133 (0.044)	0.727 (0.134)	0.267 (0.057)	0.308 (0.064)	-	-
$T_1 - T_2$ Edge	0.622 (0.036)	1.000 (0.000)	0.439 (0.037)	0.653 (0.043)	0.106 (0.023)	0.905 (0.064)	0.411 (0.037)	0.961 (0.022)	0.306 (0.034)	0.688 (0.052)	0.817 (0.029)	0.681 (0.032)
n=300												
n=300	MRGN (Truth)		MRGN (CS-q)		MRGN (CS-a)		MRPC (CS-q)		MR-GGI (CS-q)		GMAC	
	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision
M0	0.833 (0.048)	0.847 (0.047)	0.333 (0.061)	0.541 (0.082)	0.550 (0.064)	0.733 (0.066)	0.667 (0.061)	0.571 (0.059)	0.017 (0.017)	0.200 (0.179)	-	-
M1	0.817 (0.050)	0.516 (0.051)	0.617 (0.063)	0.333 (0.045)	0.567 (0.064)	0.374 (0.051)	0.133 (0.044)	0.286 (0.085)	0.517 (0.065)	0.477 (0.062)	-	-
M2	0.417 (0.064)	0.926 (0.050)	0.283 (0.058)	0.850 (0.080)	0.200 (0.052)	1.000 (0.000)	0.333 (0.061)	0.526 (0.081)	0.000 (0.000)	NA	-	-
M3	0.733 (0.057)	0.898 (0.043)	0.317 (0.060)	0.679 (0.088)	0.383 (0.063)	1.000 (0.000)	0.500 (0.065)	0.833 (0.062)	0.000 (0.000)	0.000 (0.000)	-	-
M4	0.367 (0.062)	0.846 (0.071)	0.167 (0.048)	0.345 (0.088)	0.117 (0.041)	0.389 (0.115)	0.267 (0.057)	0.571 (0.094)	0.350 (0.062)	0.304 (0.055)	-	-
$T_1 - T_2$ Edge	0.806 (0.029)	0.980 (0.012)	0.622 (0.036)	0.700 (0.036)	0.578 (0.037)	0.860 (0.032)	0.489 (0.037)	0.936 (0.025)	0.439 (0.037)	0.775 (0.041)	0.828 (0.028)	0.645 (0.031)
n=1000												
n=1000	MRGN (Truth)		MRGN (CS-q)		MRGN (CS-a)		MRPC (CS-q)		MR-GGI (CS-q)		GMAC	
	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision
M0	0.850 (0.046)	0.911 (0.038)	0.450 (0.064)	0.711 (0.074)	0.483 (0.065)	0.725 (0.071)	NA	NA	0.017 (0.017)	0.200 (0.179)	-	-
M1	0.917 (0.036)	0.632 (0.052)	0.767 (0.055)	0.422 (0.047)	0.717 (0.058)	0.417 (0.049)	NA	NA	0.667 (0.061)	0.800 (0.057)	-	-
M2	0.583 (0.064)	0.946 (0.037)	0.583 (0.064)	0.921 (0.044)	0.500 (0.065)	0.882 (0.055)	NA	NA	0.000 (0.000)	NA	-	-
M3	0.850 (0.046)	0.981 (0.019)	0.333 (0.061)	0.870 (0.070)	0.433 (0.064)	0.963 (0.036)	NA	NA	0.000 (0.000)	NA	-	-
M4	0.767 (0.055)	0.958 (0.029)	0.633 (0.062)	0.559 (0.060)	0.650 (0.062)	0.619 (0.061)	NA	NA	0.750 (0.056)	0.455 (0.050)	-	-
$T_1 - T_2$ Edge	0.917 (0.021)	0.959 (0.015)	0.844 (0.027)	0.707 (0.031)	0.817 (0.029)	0.735 (0.031)	NA	NA	0.483 (0.037)	0.737 (0.041)	0.933 (0.019)	0.672 (0.030)

295 The importance of confounding was also evident in comparisons with GMAC. In our sim-
 296 ulations, GMAC had substantially higher type I error rates for the $T_1 - T_2$ edge than previously
 297 reported by [14]. We attributed this difference to our more challenging simulation design,
 298 which included substantially more confounders and a much larger candidate pool. Repeat-
 299 ing the GMAC simulations with increasing numbers of confounders reproduced the published
 300 type I error rate when only two confounders were present, but showed progressively inflated
 301 type I error as the number of confounders and candidate pool size increased (Supplementary
 302 Figure 8).

303 We further examined the impact of other types of confounding on MRGN inference in
 304 Simulation B. Including an intermediate among the selected confounders had relatively little
 305 effect on either causal-model recovery or $T_1 - T_2$ edge detection (Figure 3; Supplementary
 306 Table 2). In particular, although adjustment for an intermediate could potentially attenuate the
 307 association between T_1 and T_2 , we did not observe a substantial reduction in $T_1 - T_2$ edge
 308 recall. In contrast, the common child was incorrectly selected as a confounder in 66% of the
 309 trios, and including this variable substantially reduced model recall for M0 and M3. This is
 310 expected because of the association between the two parent nodes induced by the common
 311 child, leading to a false positive $T_1 - T_2$ edge. Nevertheless, MRGN maintained reasonable
 312 recall for the $T_1 - T_2$ edge even when both an intermediate and a common child were present.

313 3.2 Application to GTEx data

314 The GTEx consortium has identified numerous expression quantitative trait loci (eQTLs) as-
 315 sociated with gene expression across human tissues [26]. Many eQTLs are associated with
 316 both cis- and trans-genes, motivating studies of whether cis-genes mediate their effects on
 317 trans-genes [14]. We previously applied MRPC across GTEx tissues to identify diverse causal
 318 structures among eQTLs and their associated cis- and trans-genes, including protein-coding
 319 genes and long non-coding RNAs (lncRNAs) [20]. Here, we use MRGN to further explore

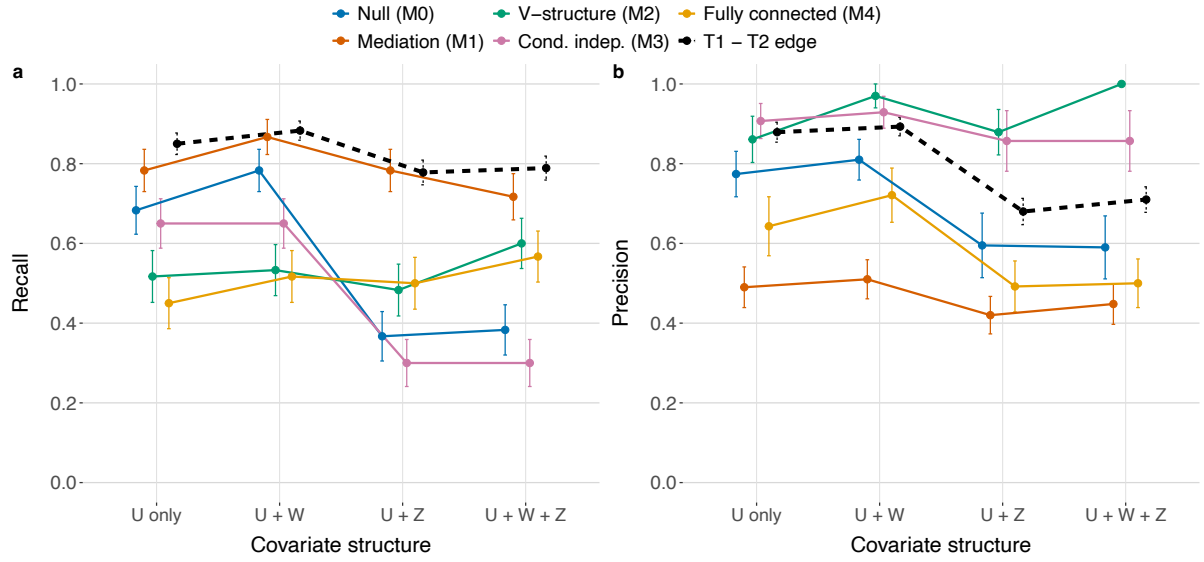


Figure 3: Recall and precision of MRGN with CS-q in Simulation B under different covariate structures. (a) Recall and (b) precision for the five causal models and the $T_1 - T_2$ edge, aggregated across effect sizes. U , W , and Z denote multiple confounders, an intermediate variable, and a common child, respectively. Dots show mean recall or precision, and error bars show one standard error.

Alt text: Two panels compare recall and precision across covariate structures. The left panel shows that recall varies substantially by causal model and decreases most notably with a common child, whereas the right panel shows that precision generally is higher and varies less.

320 these regulatory relationships in GTEx whole blood. Because the trios are defined by prior
321 eQTL-gene associations, the inferred structures are interpreted as exploratory causal models
322 conditional on this selection.

323 We applied the preprocessing and normalization pipeline in our previous study to whole
324 blood data from 670 GTEx v8 samples [20]. Gene expression ~~was PEER-normalized [29]~~
325 values were residuals from PEER normalization [29], with all GTEx-reported covariates included
326 as known covariates: the top five genotype-derived principal components (PCs) to account for
327 population structure, sex, age, PCR batch, and platform. Each trio consisted of an eQTL and its
328 cis- and trans-target genes of highest association, where cis-genes were located within 1 Mb of
329 the eQTL and trans-genes were located more than 1 Mb away or on a different chromosome.
330 After preprocessing, 8,823 trios were identified. Restricting the analysis to protein-coding
331 genes and long non-coding RNAs (lncRNAs) to reduce transcript mismapping yielded a final
332 set of 3,248 trios (Supplementary Table 3). When comparing our method to GMAC, we im-
333 puted missing genotypes using multiple correspondence analysis implemented in the *missMDA*
334 R package [30], since GMAC cannot accommodate missing genotype values.

335 Since the sample size 670 is much smaller than the number of genes (around 56,000) in the
336 genomewide expression data, principal component analysis (PCA) of the PEER-normalized
337 expression data yielded 670 PCs (with the last PC being essentially zero) as candidate con-
338 founders. Using CS-q with $q < 0.05$, MRGN selected on average 30 confounders per trio
339 (SD = 6.79) (Figure 4a). Using CS- α with $\alpha < 0.01$, MRGN selected on average 51 con-
340 founders per trio (SD = 6.38) (Supplementary Figure 9). These PCs represent potential effects
341 of genes outside the gene pair in each trio. Although the PCs were derived from all genes,
342 excluding the two genes in a given trio would have negligible impact on the resulting PCs.
343 ~~Covariates provided by GTEx, namely sex, platform, and PCR batch, were also included as~~
344 ~~known confounders for each trio.~~

MRGN identified more mediation and conditional independence trios

Following confounder selection with CS-q, MRGN identified ~~712~~715 trios as the null model, ~~76~~79 as mediation, ~~11~~9 as v-structure, ~~2,278~~2,268 as conditionally independent, ~~152~~158 as fully connected, and 19 as “other” (Figure 4b). With CS- α , these numbers are 911 null, 96 mediation, 10 v-structure, 2,063 conditionally independent, 111 fully connected, and 57 “other” (Supplementary Figure 9). Focusing on the results from CS-q, MRGN identified twice as many mediation trios and 389 additional conditional independence trios than MRPC [20], although the total number of inferred $T_1 - T_2$ edges was similar across methods (Figure 4c). As in MRPC, MRGN distinguished two mediation models (Figure 4d): cis-gene mediation ($V \rightarrow T_1 \rightarrow T_2$) and trans-gene mediation ($V \rightarrow T_2 \rightarrow T_1$). Among the 76 mediation trios, 43 (57%) involved cis-gene mediation and 33 (43%) involved trans-gene mediation (Figure 4d).

Nine cis-gene mediation trios and 11 trans-gene mediation trios were identified by both methods, suggesting substantial differences in the inferred causal models. Examination of trios classified as mediation by MRGN but not MRPC indicated that these discrepancies were primarily driven by differences in confounder selection. For example, for the trio consisting of rs117667029, METTL26, and PTMS, MRGN identified 37 confounders and inferred mediation ($\text{rs117667029} \rightarrow \text{METTL26} \rightarrow \text{PTMS}$), whereas MRPC identified only three confounders and inferred conditional independence ($\text{METTL26} \leftarrow \text{rs117667029} \rightarrow \text{PTMS}$). Repeating the MRGN analysis with the same three confounders also yielded conditional independence, confirming that accurate confounder selection is critical for consistent causal network inference.

To assess the robustness and uncertainty of the inferred mediation trios, we applied baycn [32], a Bayesian DAG method that estimates posterior probabilities for edge presence and direction under the PMR constraint, to each trio together with its selected confounders (Supplementary Table 4). Unlike MRGN, baycn treats the selected confounders as network nodes and infers their edge directions rather than assuming that they point to the genes. We ran baycn for ~~1~~4 million iterations with 20% burn-in and thinning every ~~28~~28,000 iterations. An edge is

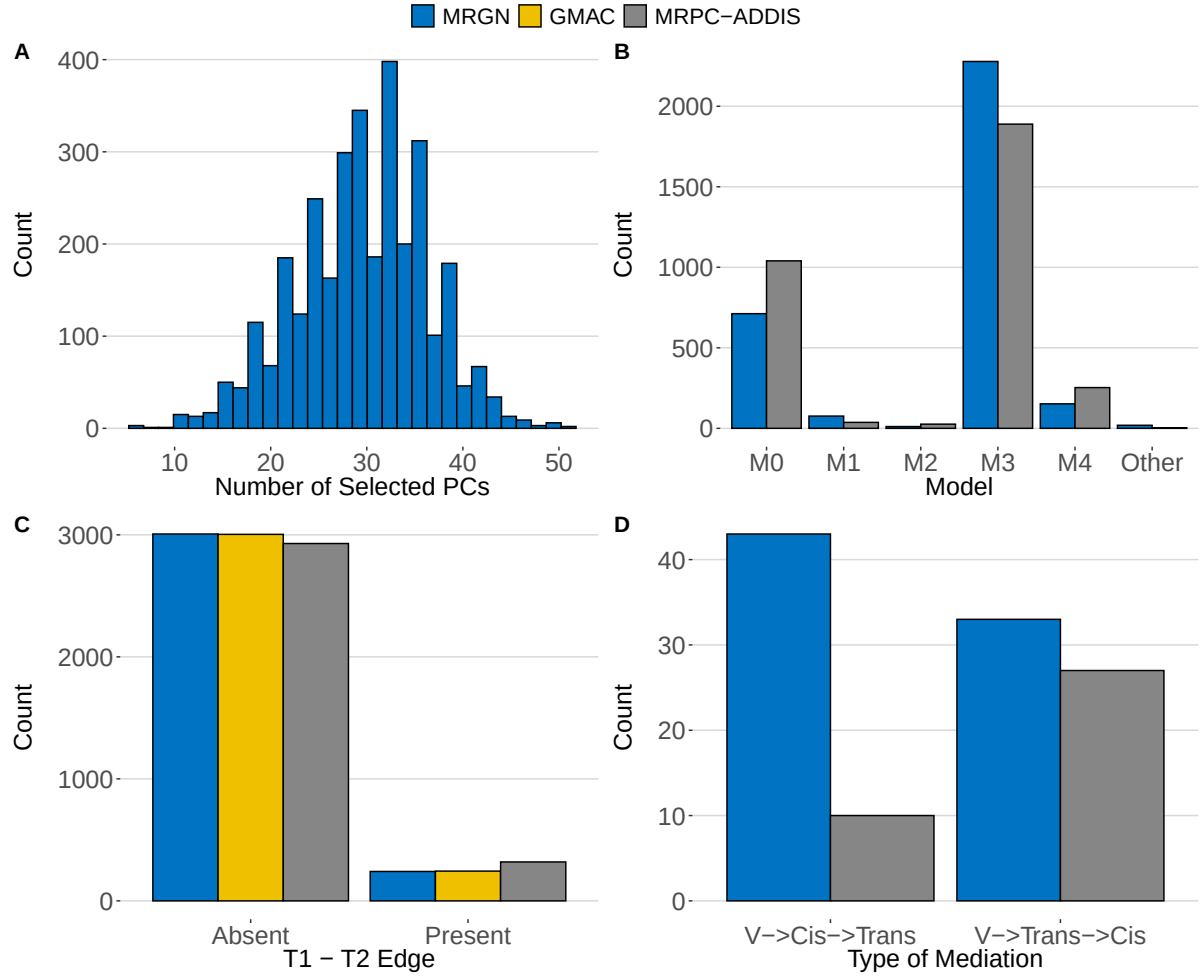


Figure 4: Comparative analysis of MRGN with CS-q, MRPC [31] and GMAC [14] on GTEx human whole blood trios. We applied MRGN with CS-q, GMAC, and MRPC (with ADDIS for FDR control) to genomic trios. (a) Histogram of the number of Principal Components (PCs) selected as confounders per trio. (b) Count of inferred causal models (M1–M4 and Other) comparing MRGN and MRPC. (c) Counts of trios with the $T_1 - T_2$ edge inferred as absent or present by the three methods. (d) Counts of two types of mediation models (via cis-gene vs. trans-gene) inferred by MRGN and GMAC.

Alt text: Four panels show that selected confounder counts are concentrated around 30, MRGN and MRPC differ in counts of inferred causal models, all three methods infer similar numbers of $T_1 - T_2$ edges, and counts of two types of mediation models differ between MRGN and GMAC.

considered presence if the edge absence probability is < 0.5 , and an edge is considered directed if the absolute difference between the probabilities for the two directions is > 0.2 . Among the 43 cis-gene mediation trios, baycn inferred ~~17~~(~~40~~15 (35%)) as the same mediation model, ~~25~~(~~58~~28 (65%)) as M4, and one as M3. Among the ~~33~~36 trans-gene mediation trios, ~~8~~(~~24~~9 (25%)) were inferred as the same mediation model and ~~19~~(~~58~~23 (64%)) as M4, with the remaining six assigned to other models. Considering only the directed $T_1 - T_2$ edge, the two methods agreed for ~~28~~(~~65~~30 (70%)) cis-gene and ~~21~~(~~64~~24 (67%)) trans-gene mediation trios. Thus, the predominant discrepancy was between M1 and M4, consistent with the M1-M4 confusion observed in the simulations.

MRGN is robust to model misspecification in inference of mediation

We further investigated the effect of inference model misspecification on mediation inference using 192 mediation trios identified by GMAC in whole blood. Four scenarios were considered, ranging from models using the true confounders to models omitting many moderate-to-strong confounders (see Supplementary Text 1).

Recall remained around 93% when the inference model included most influential confounders, even if a few weak confounders were incorrectly included or omitted. In contrast, omitting many moderate-to-strong confounders reduced recall to approximately 70%, demonstrating that retaining influential confounders is critical for accurate causal network inference.

Biological interpretation

We formulated genomic trios to discover possible causal relationships among genes selected for their strong associations with the eQTL, rather than known regulatory relationships. Most selected genes are not transcription factors (TFs), and many are relatively understudied lncRNAs. Accordingly, only four out of 74 unique mediator genes are TFs represented in CollecTRI [33], a curated database of high-confidence TF-target interactions, and none of the gene pairs

395 in the inferred mediation trios are reported TF-target pairs. Nevertheless, literature evidence
396 supports or suggests biological mechanisms for several inferred relationships:

- 397 • chr1:25253604_G_A → TMEM14C (trans-gene on chr6) → RHCE (cis-gene), with
398 Pearson correlations $r_{V,cis} = -0.05$, $r_{V,trans} = -0.22$, and $r_{cis,trans} = 0.16$. Both genes are
399 involved in erythroid maturation. TMEM14C, a mitochondrial inner-membrane trans-
400 porter, controls intracellular heme levels [34]; heme regulates GATA1 activity and stabil-
401 ity [35], and GATA1 in turn transactivates RHCE [36, 37]. The heme-GATA1 feedback
402 loop is well established in erythroid biology [38], providing a plausible mechanism for
403 the inferred TMEM14C→RHCE relationship.
- 404 • chr5:122044080_C_T → EFEMP1 (trans-gene on chr2) → LOX (cis-gene), with $r_{V,cis} =$
405 0.03 , $r_{V,trans} = 0.19$, and $r_{cis,trans} = 0.39$. EFEMP1 paralog EFEMP2 (a.k.a. fibulin-4
406 or FBLN4) and FBLN5 (a.k.a. fibulin-5) have direct mechanistic relationships with the
407 LOX family: EFEMP2 binds LOX and recruits it for elastin crosslinking, while FBLN5
408 interacts with several LOX-family proteins [39, 40]. As EFEMP1 functions in the same
409 extracellular-matrix assembly module, the inferred edge suggests a potential role for
410 EFEMP1 in LOX regulation.
- 411 • chr17:19867907_C_T → ULK2 (cis-gene) → GSDMC (trans-gene on chr8), with $r_{V,cis} =$
412 -0.11 , $r_{V,trans} = 0.18$, and $r_{cis,trans} = -0.12$. ULK1, a close paralog of ULK2, di-
413 rectly phosphorylates and activates GSDMA, a paralog of GSDMC, during starvation-
414 induced pyroptosis [41]. Gasdermins including GSDMA and GSDMC are also linked
415 to autophagy [42], while ULK2 is a core autophagy-initiating kinase. The inferred
416 ULK2→GSDMC edge therefore suggests a potential interaction between these two gene
417 families.

4 Discussion

In summary, we have developed MRGN to improve the power and accuracy of detecting key causal relationships in diverse genomic trios, particularly in the presence of multiple confounding variables. Our method adopts an extended interpretation of the PMR [8] and prioritizes conditional independence tests to identify five basic causal trio structures. A major advantage of our approach is the reduction of the multiple testing burden; by adjusting for selected confounders within a regression framework, MRGN can incorporate a large number of confounders without the need to infer a graph larger than three nodes.

A general challenge in causal structure learning is that statistical support for individual edges is continuous, whereas interpreting the inferred structure often requires deciding whether an edge is present and, if so, its direction. Constraint-based methods make these discrete decisions through thresholds in hypothesis testing, while score-based and Bayesian methods may similarly require thresholds on edge weights or edge inclusion probabilities. Thus, converting continuous statistical evidence into discrete edge decisions generally requires a pre-specified criterion. In constraint-based methods, failure to reject the null hypothesis may indicate either genuine edge absence or insufficient statistical support for edge presence. This leads to difficulty distinguishing models, particularly M1 and M4, which differ by a single conditional independence test, as we observe in the simulations and GTEx analysis. As an alternative, equivalence testing may provide stronger evidence for edge absence, but requires specifying the equivalence bounds, which may depend on the phenotype, application, and scale of the data. Thus, equivalence testing does not eliminate the need for a threshold, but instead requires a threshold to be specified for a different quantity.

While our simulations and real data analysis demonstrate the flexibility of MRGN in estimating causal relationships, the method is currently limited to trios containing a genetic variant that acts as an instrumental variable. In larger networks, candidate gene pairs may first be iden-

tified based on biological interest, observed associations, or other prior information. Genetic variants associated with one or both genes can then be identified and included as anchors for causal inference, as in our GTEx analysis. Overlapping trios may produce relationships that are not globally consistent, and assembling them into a coherent network requires additional methodology. We are currently extending MRGN to a general method that jointly infers larger networks and incorporates potential intermediate variables and common children as additional network nodes, while confounders are accounted for in the inference as in MRGN.

Data Availability

Data from The Genotype-Tissue Expression (GTEx) Project version 8 were obtained from dbGaP (<http://www.ncbi.nlm.nih.gov/gap>) through the following accession number: phs000424.v8.pht002741.v8.p2. Open-source R packages used in this manuscript include MRGN; <https://github.com/audreyfulab/MRGN>, MRPC (version 3.1.0); <https://cran.r-project.org/web/packages/MRPC/index.html>, and GMAC (version 3.1); <https://cran.r-project.org/web/packages/GMAC/index.html>. R code for simulations and applications with results is available at https://github.com/audreyfulab/MRGN_Manuscript_Repo. Additional results are also included in the supplementary material.

Conflicts of Interest

The authors declare no competing interests.

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485 www.ncbi.nlm.nih.gov/gap through dbGaP accession number phs000424.v8.pht002741.v8.p2.

486 **Author Contributions**

487 J.K and A.Q.F designed the study, J.K developed the tools and analyzed the data, and J.K and
488 A.Q.F wrote the manuscript.

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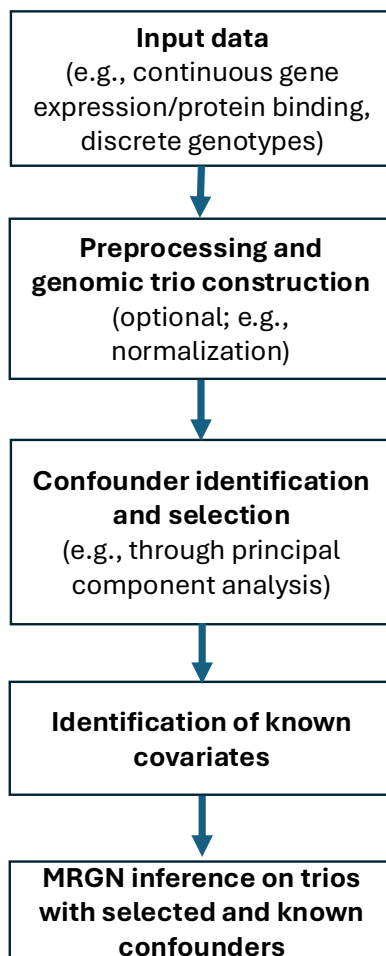
Biographical Note

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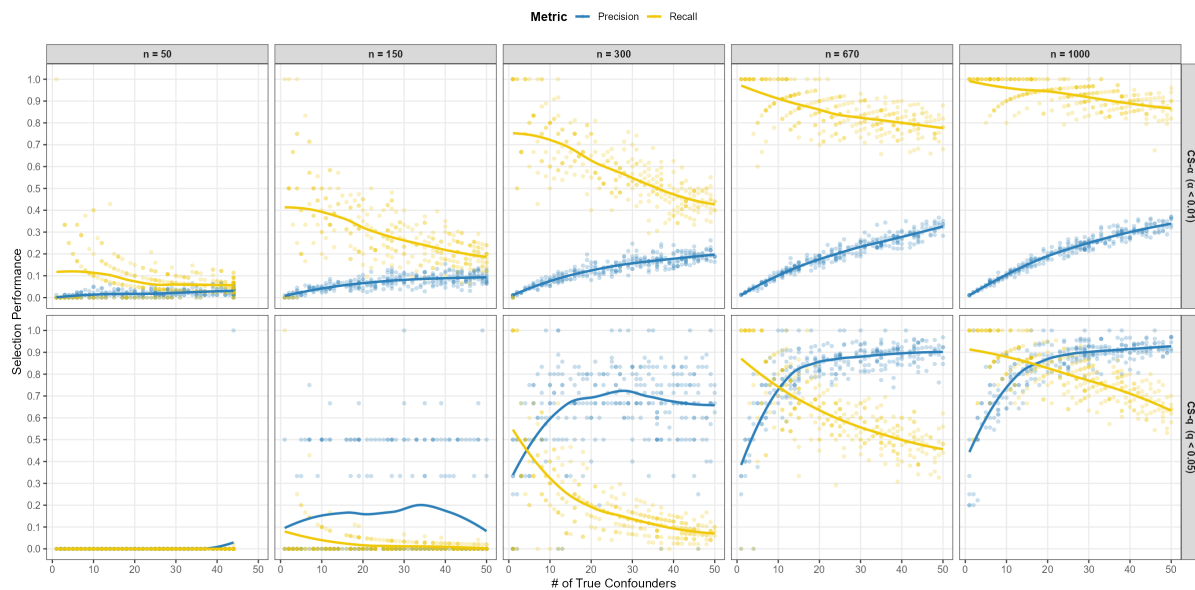
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621 the Wayne State University School of Medicine, Detroit, Michigan, USA. She is also a faculty
622 member of the Institute for AI and Data Science at Wayne State University.

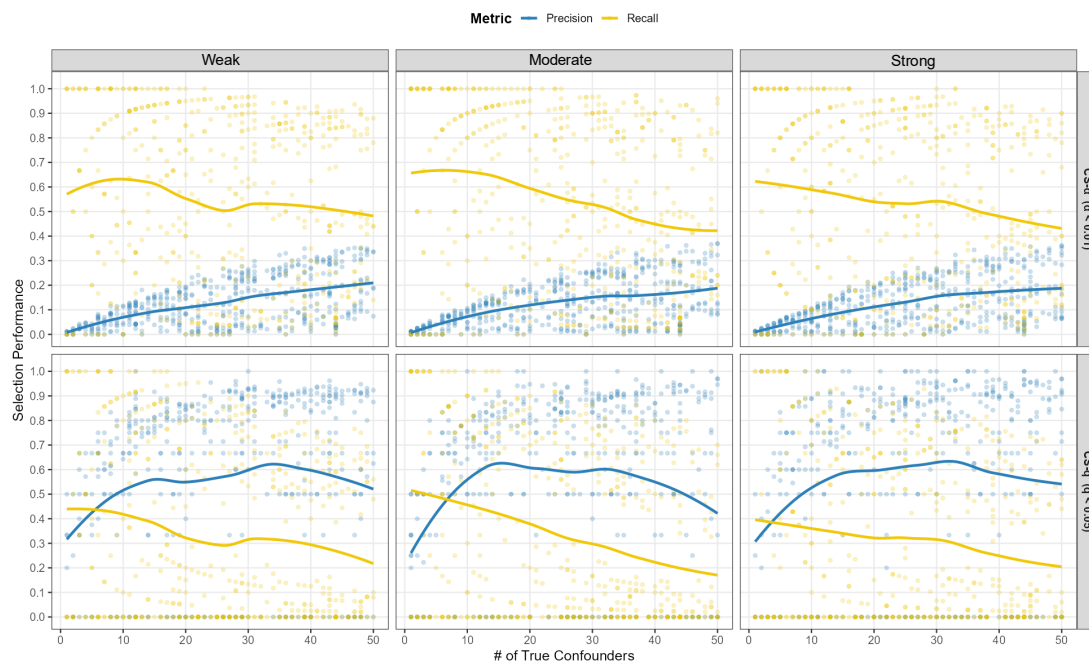
623 **Supplementary figures**



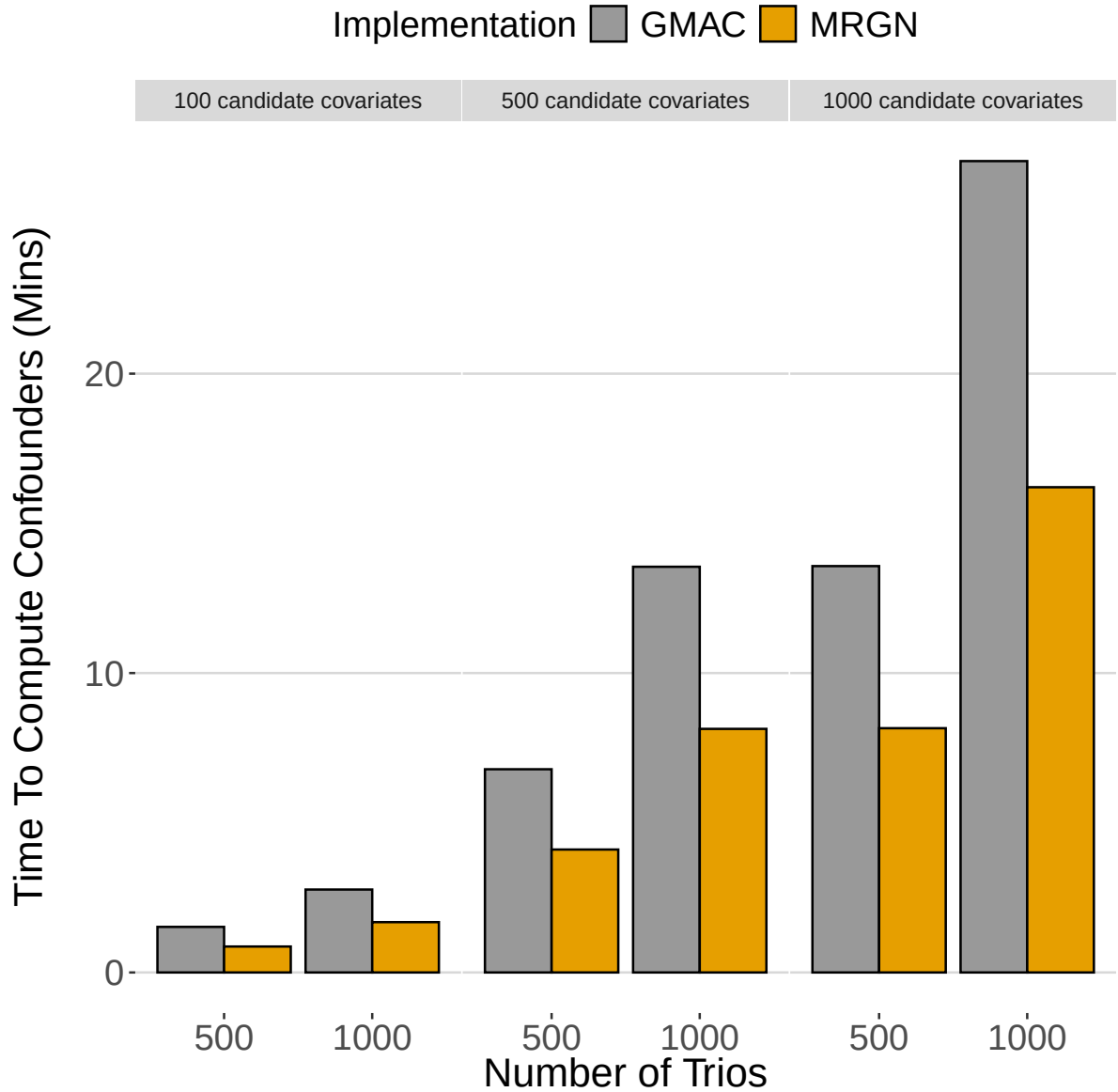
Supplementary Figure 1: The flowchart of using MRGN for data analysis.



Supplementary Figure 2: Precision and recall of confounder selection on simulated data by two multiple-testing correction methods: CS-q and CS- α , stratified by sample size. Each dot is a simulated data set.



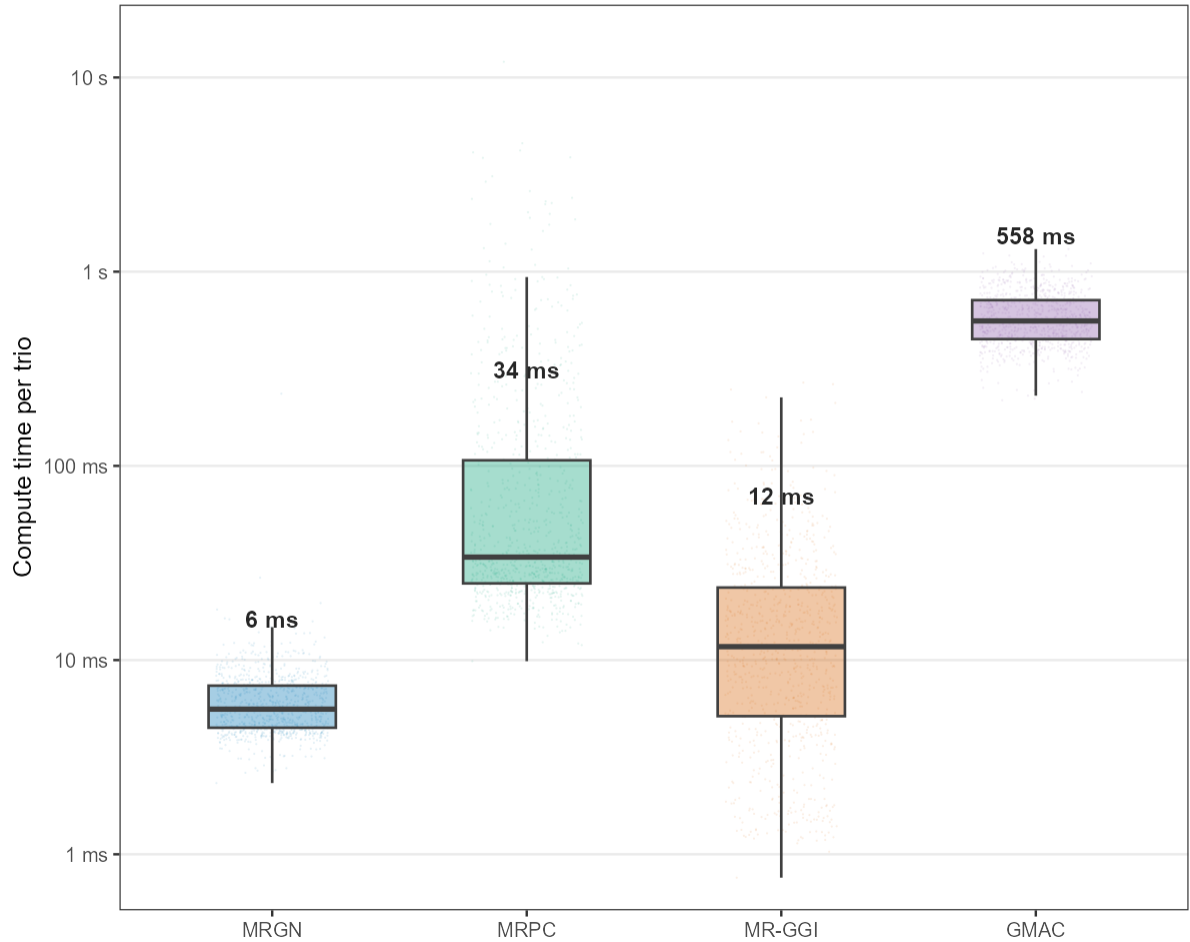
Supplementary Figure 3: Precision and recall of confounder selection on simulated data by two multiple-testing correction methods: CS-q and CS- α , stratified by effect size. Each dot is a simulated data set.



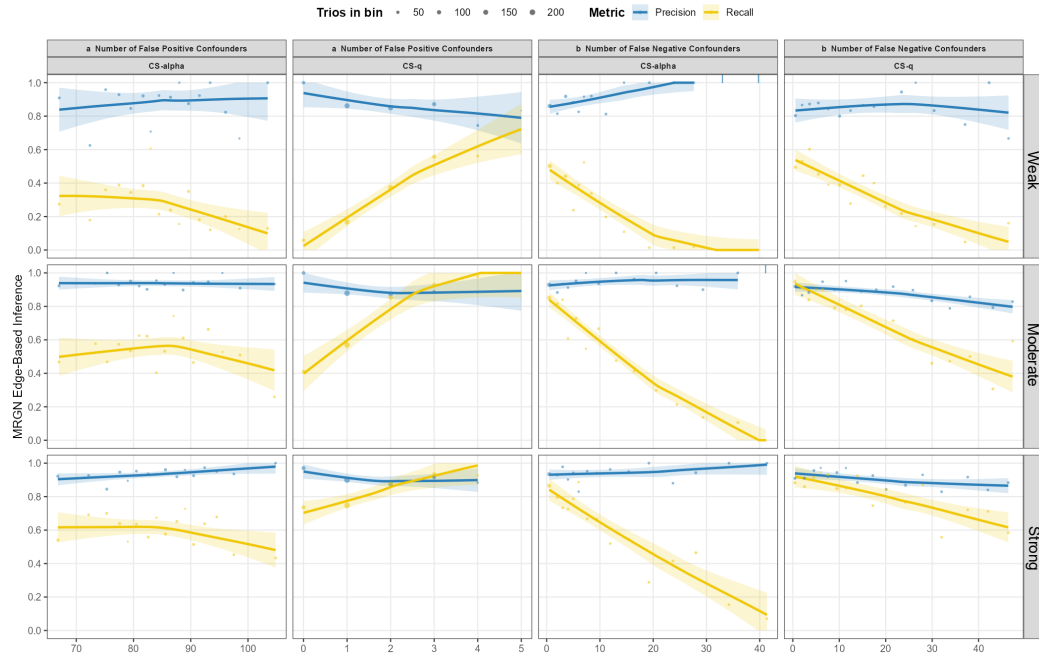
Supplementary Figure 4: Comparison of computation times between our implementation of the CS-q method for confounder selection and that of the GMAC method [14]. Computation times are shown across different numbers of input trios and sizes of the candidate confounder pool.

Per-trio compute time, as deployed

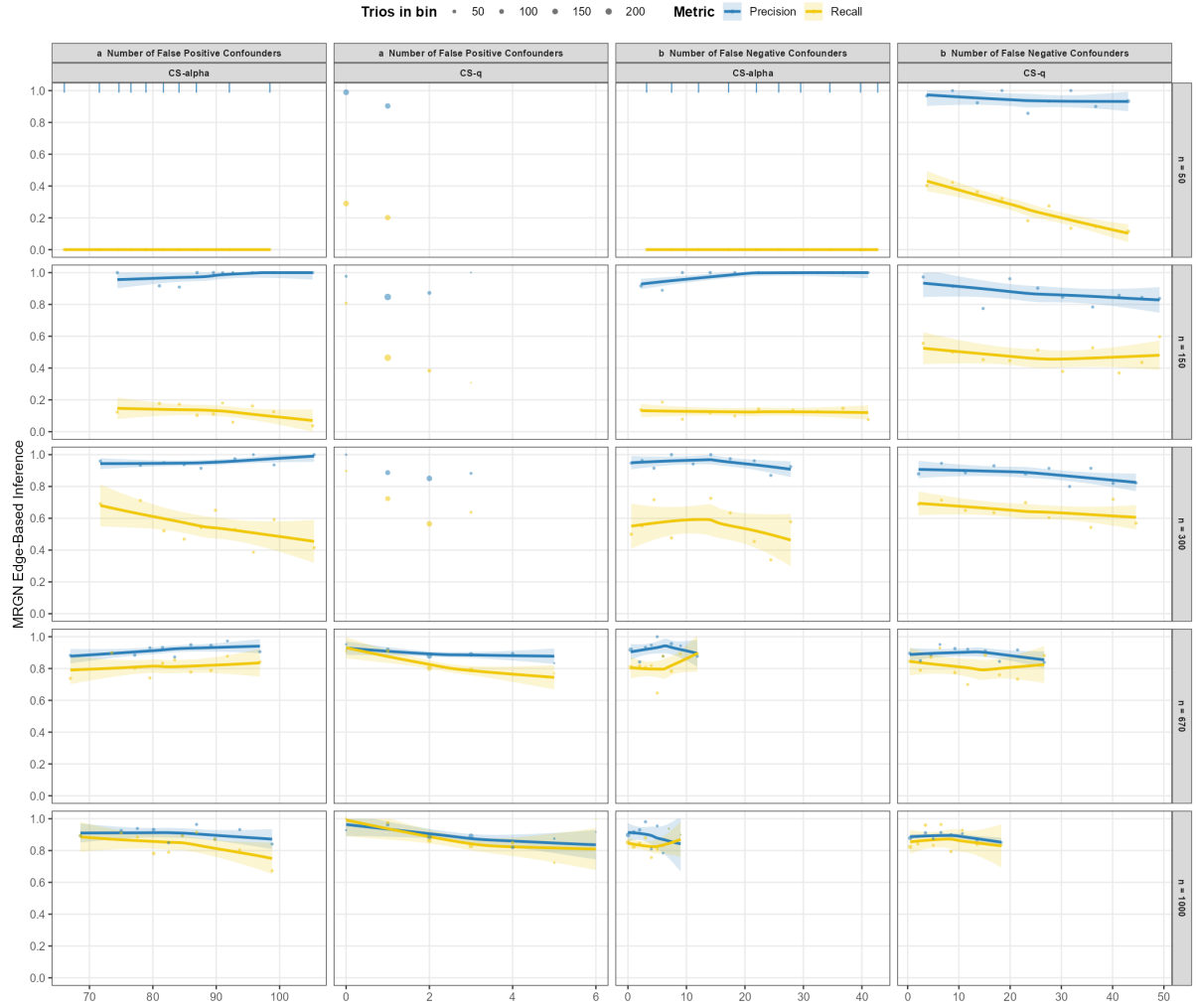
n = 50, 150, 300 -- the 900 trios all four methods cover. Box = IQR, whiskers = 1.5 x IQR, label = median.
MRGN / MRPC / MR-GGI on the CS-q confounder set, GMAC on its own selection. Log scale.



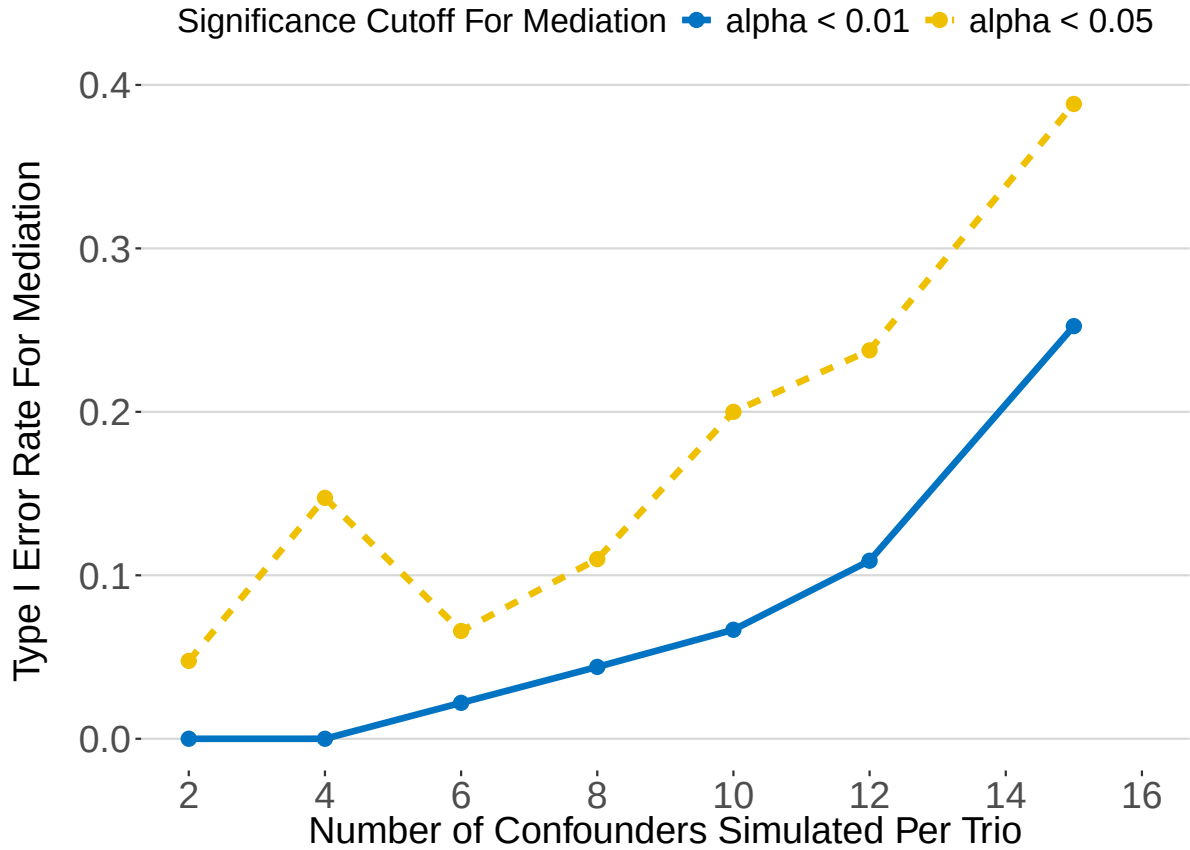
Supplementary Figure 5: Boxplots of runtime for causal network inference after confounder selection from different methods on simulated trios.



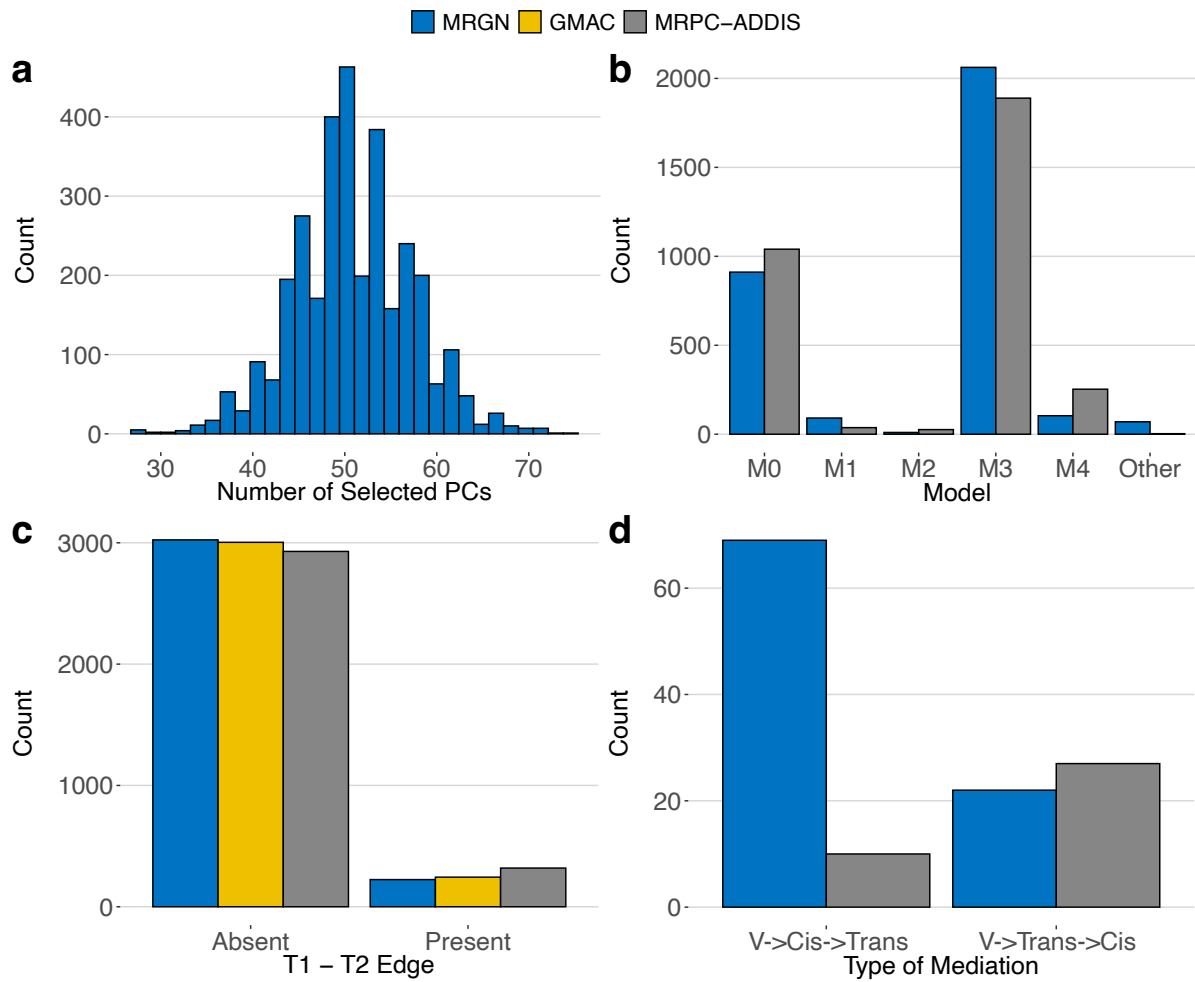
Supplementary Figure 6: Relationship between confounder selection and $T_1 - T_2$ edge inference, stratified by false positive/negative confounders and effect size.



Supplementary Figure 7: Relationship between confounder selection and $T_1 - T_2$ edge inference, stratified by false positive/negative confounders and sample size.



Supplementary Figure 8: Comparison of GMAC and MRGN type I error rates for $T_1 - T_2$ detection under confounding. Results of simulations investigating type I error rate (false mediation) under the null model. Trios were simulated under the null model for each of seven scenarios: 2, 4, 6, 8, 10, 12 or 15 confounders, excluding common child and intermediate variables. The GMAC method was applied to trios in each scenario using CS-q confounder selection.



Supplementary Figure 9: Comparative analysis of MRGN with CS- α , MRPC [31] and GMAC [14] on GTEx human whole blood trios.

624 **Supplementary tables**

Supplementary Table 1: Confusion matrix aggregated over effect sizes and sample sizes in Simulation A for MRGN and MRPC. Confounder selection CS-q was used. Each method is displayed on a separate sheet.

Supplementary Table 2: Recall and precision aggregated over effect sizes in Simulation B.

Supplementary Table 3: Inference results from MRGN, MRPC, and GMAC for 3,248 whole blood trios from GTEx v8. All analyzed GTEx whole-blood trios, including genetic annotations (locus, variant ID, gene name, etc.) and inference results from MRGN, MRPC-ADDIS, and GMAC.

Supplementary Table 4: Inference results from baycn on MRGN-inferred mediation trios.

625 **Supplementary text**

Supplementary Text 1: Details of the four simulation scenarios in section “MRGN is robust to model misspecification for inference of mediation”.